



ITI Cmos Analog IC Design

Lab 09

Two stage OTA miller compensation

Mini project 1

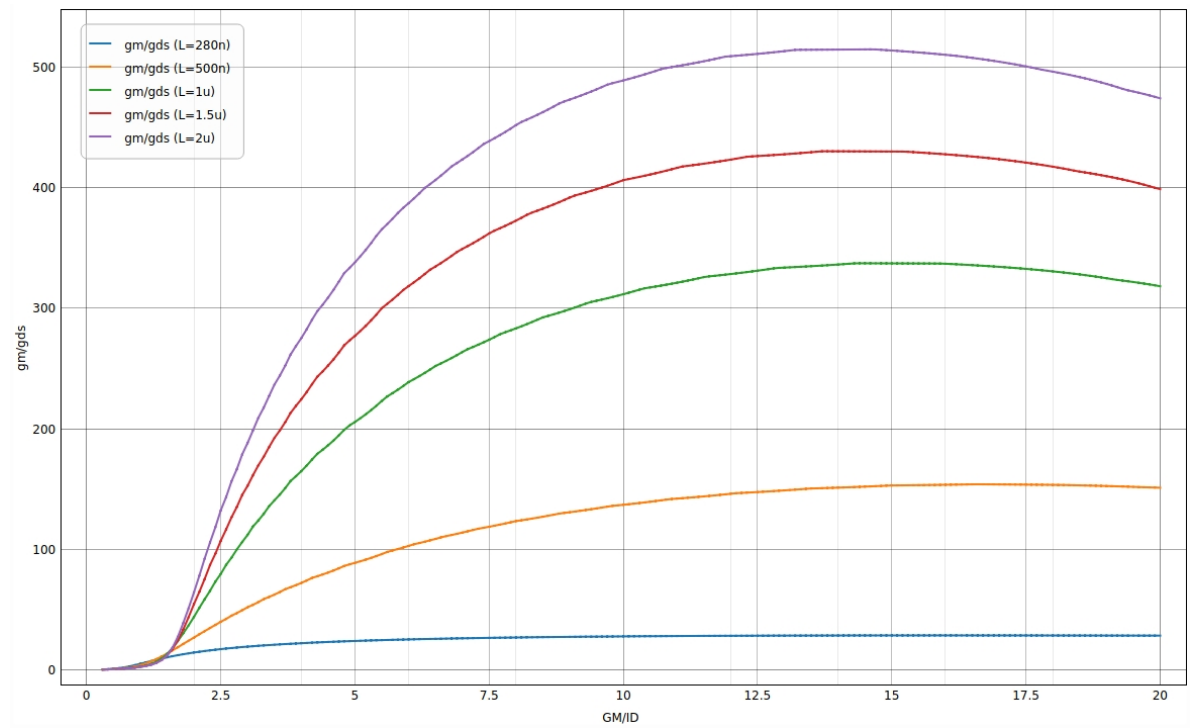
Student name: Mohamed Adel Ali

Section 7 online

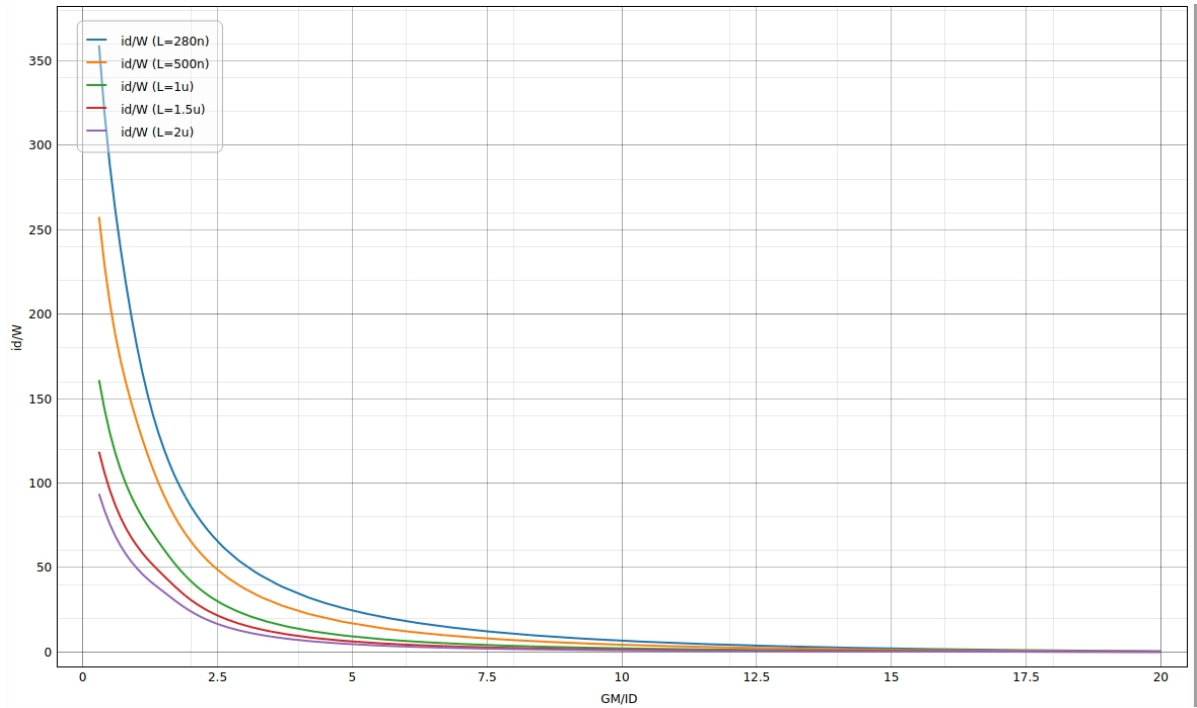
Part 1 — gm/ID Design Charts

NMOS charts

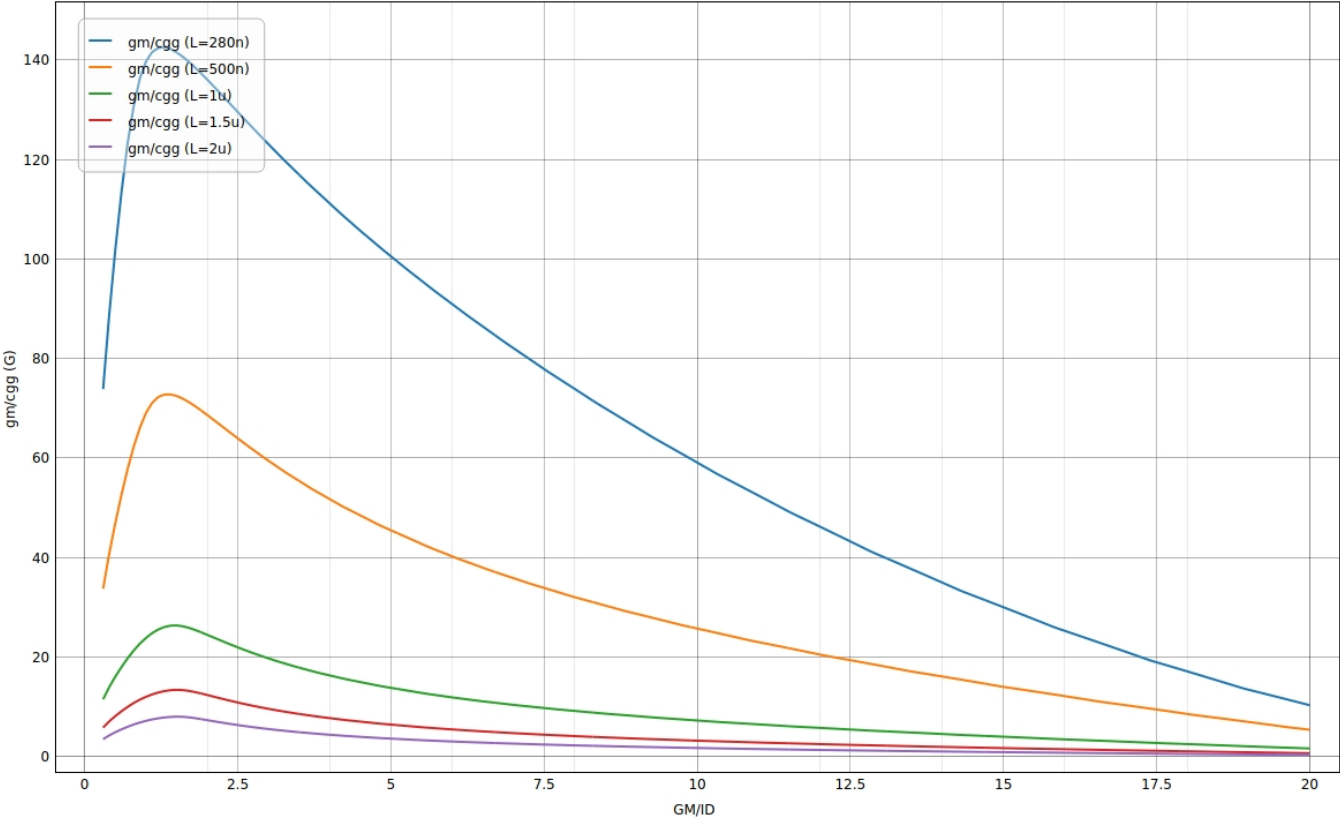
gm/gds vs gm/id



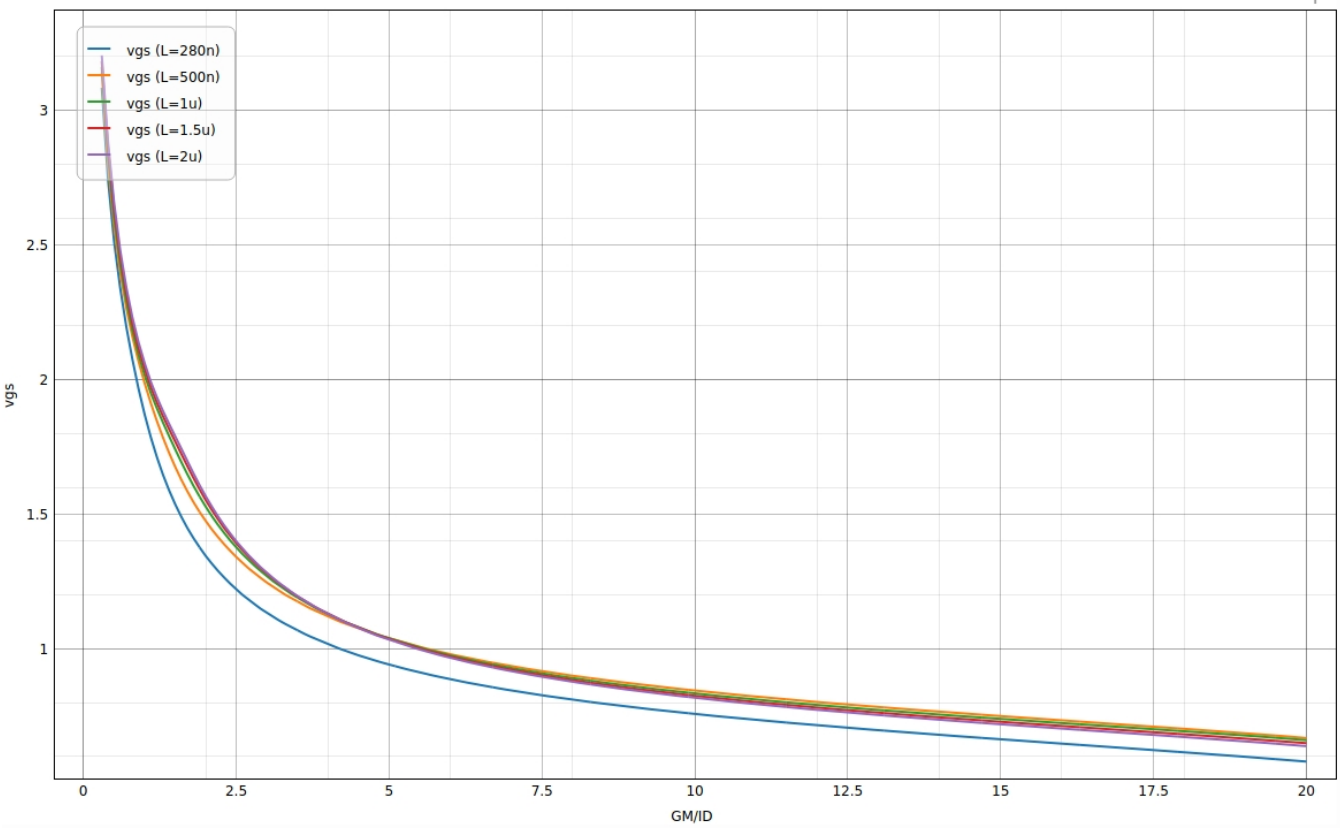
id/W vs gm/id



gm/Cgg vs gm/id

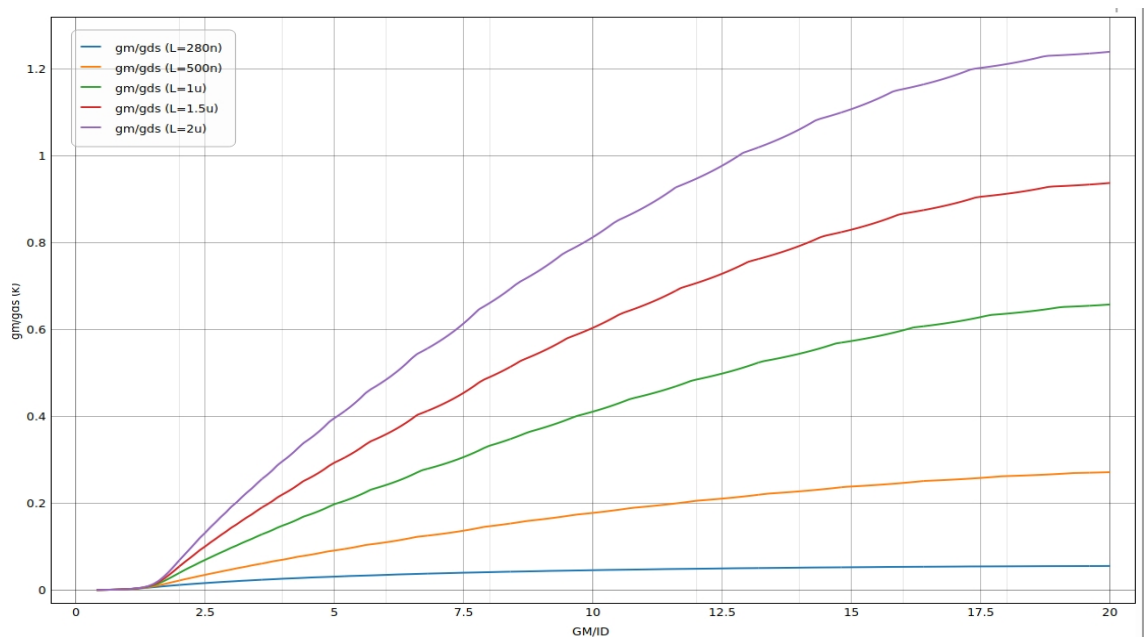


Vgs vs gm/id

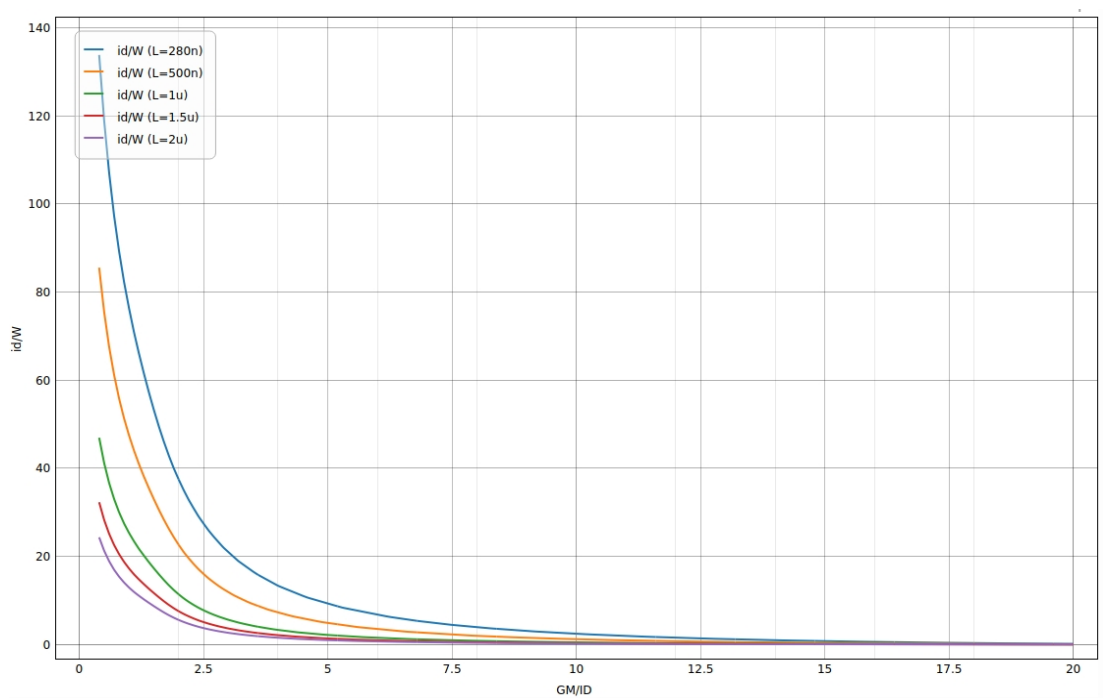


PMOS charts

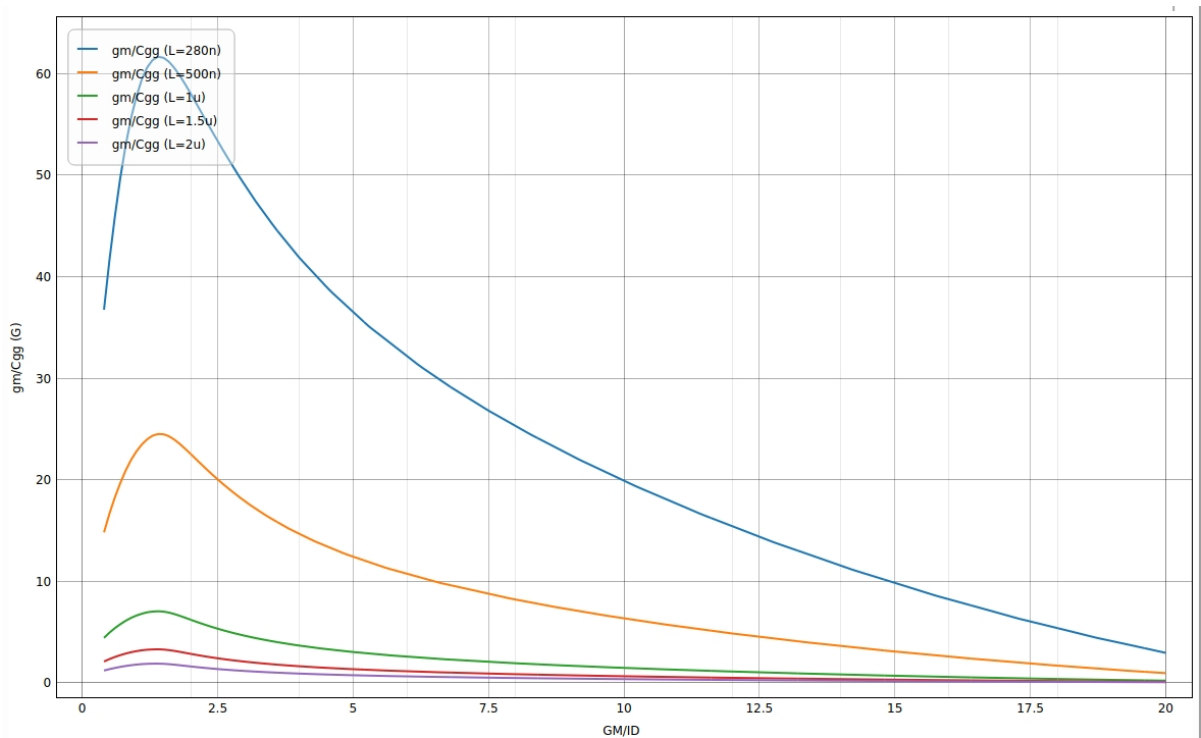
gm/gds vs gm/id



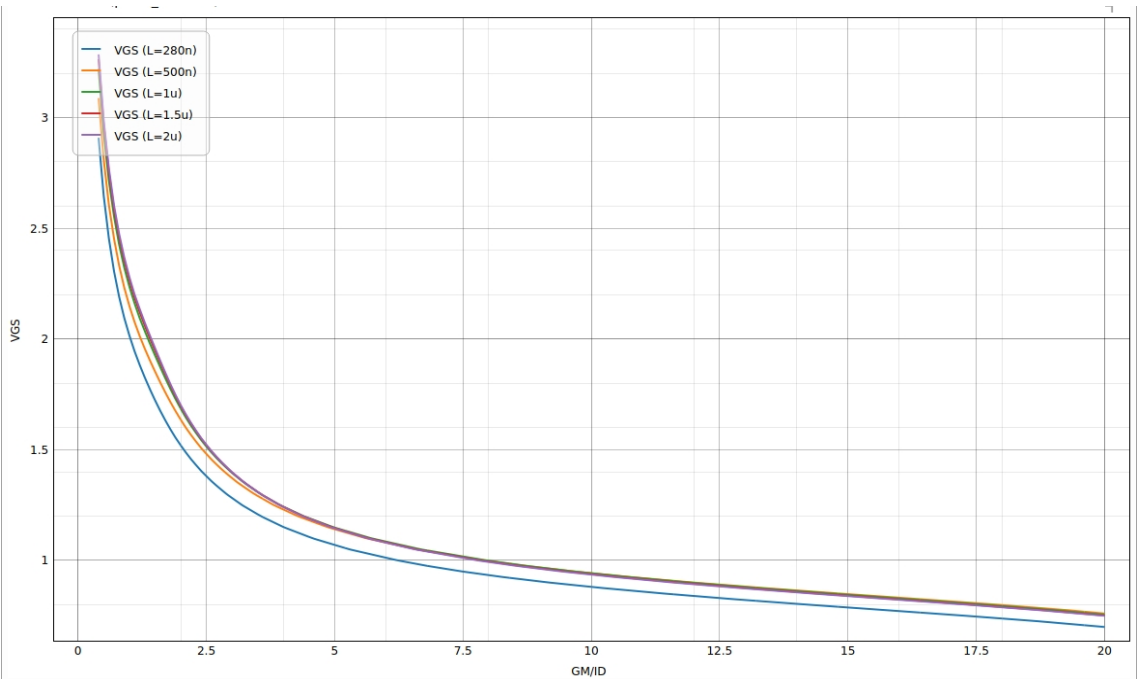
id/W vs gm/id



gm/Cgg vs gm/id



Vgs vs gm/id



Part 2 — OTA Design

Design Specifications (0.18 μm column)

Specification	Value
VDD	1.8 V
CL	5 pF
Rise time (10%–90%)	≤ 70 ns
Slew Rate	5 V/ μs
Gain error (closed-loop)	$\leq 0.05\%$
CMRR @ DC	≥ 74 dB
Phase Margin	$\geq 70^\circ$
Current budget	≤ 60 μA
CMIR-high (target from professor)	0.8 V
Swing high/low	0.2 – 1.6 V

I. Input Differential Pair (M7 / M8)

Input pair will be PMOS since CMIR is close to GND

Step 1) Choose an initial value for C_c :

$$C_c = 0.5 \cdot C_L = 0.5(5 \text{ pF}) = 2.5 \text{ pF}$$

Step 2) Compute UGF from the rise-time:

$$t_{\text{rise}} = 2.2\tau \Rightarrow \omega_u \approx f_u = 5 \text{ MHz}$$

Step 3) Compute $g_{m1,2}$ from the UGF:

$$\omega_u \approx g_m / (2\pi C_c) \Rightarrow g_{m7,8} \approx 78.6 \mu\text{S} \approx 79 \mu\text{S}$$

Step 4) Compute I_{B1} , I_{B2} from the Slew Rate:

$$SR = I_{B1} / C_c \Rightarrow I_{B1} = 12.5 \mu\text{A} \Rightarrow I_{D7,8} = 6.25 \mu\text{A} \\ I_{B2} = 60 - 12.5 = 47.5 \mu\text{A}$$

Step 5) Compute g_m/I_D for the input pair:

$$(g_m/I_D)_{7,8} = 78.6 / 6.25 \approx 12.8$$

Step 6) DC Gain from the gain-error specification:

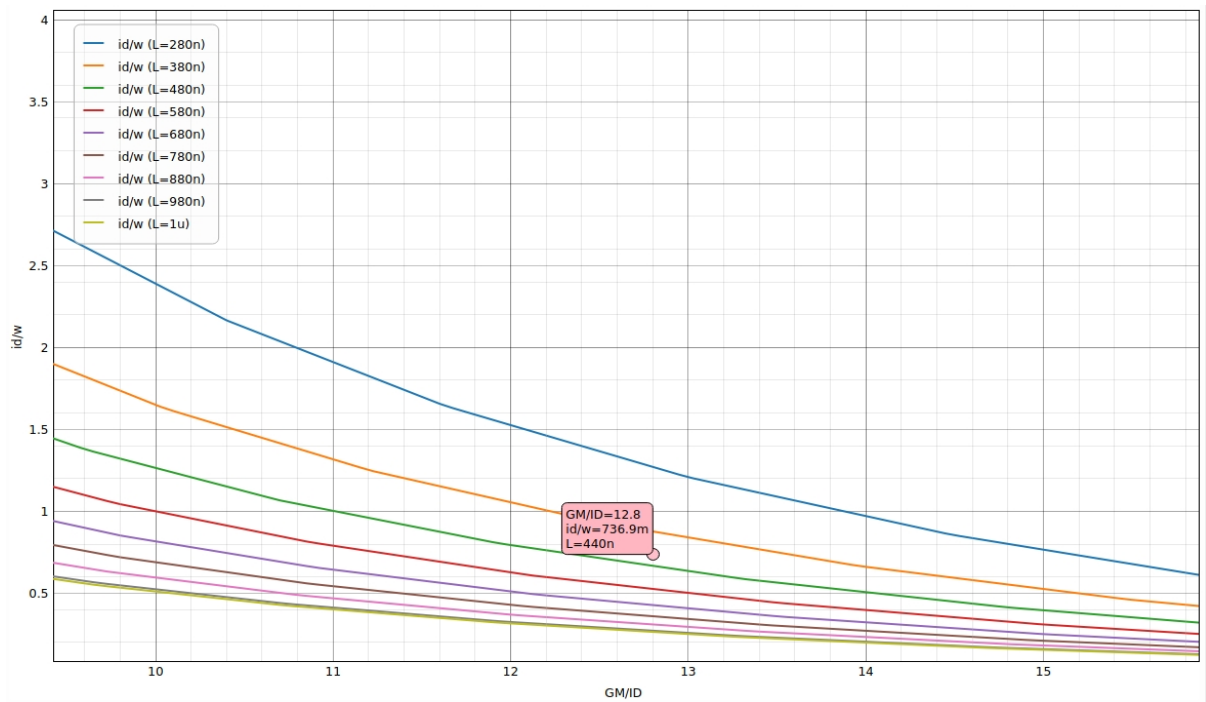
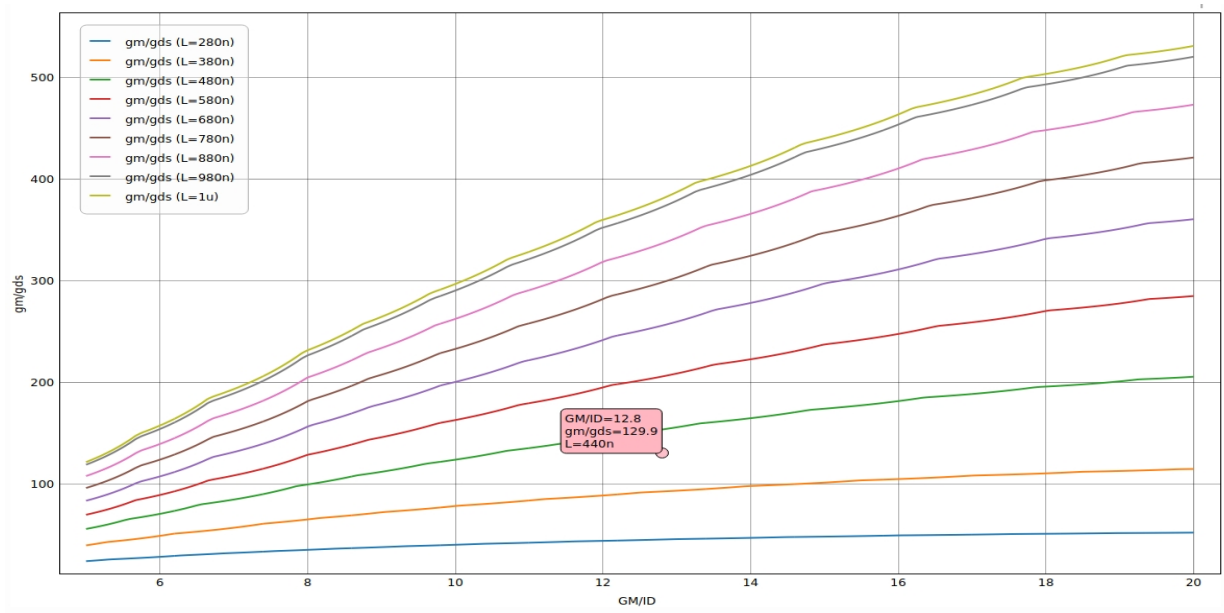
$$A_v \geq 66 \text{ dB} \approx 2000 \text{ (V/V)}$$

Step 7) gain between the two stages (6 dB split):

$$A_{v1} = 63.2 \text{ (36.0 dB)}, A_{v2} = 31.6 \text{ (30.0 dB)}$$

Step 8) Compute L for the input transistor:

$$(g_m/g_{ds})_7 > 126.2$$



From the charts: $L = 440 \text{ nm}$

Compute W :

$$W_{7,8} = I_D / (I_D/W)_{\text{chart}} = 6.25 \mu\text{A} / 773.3 \text{ m} \approx 8.58 \mu\text{m}$$

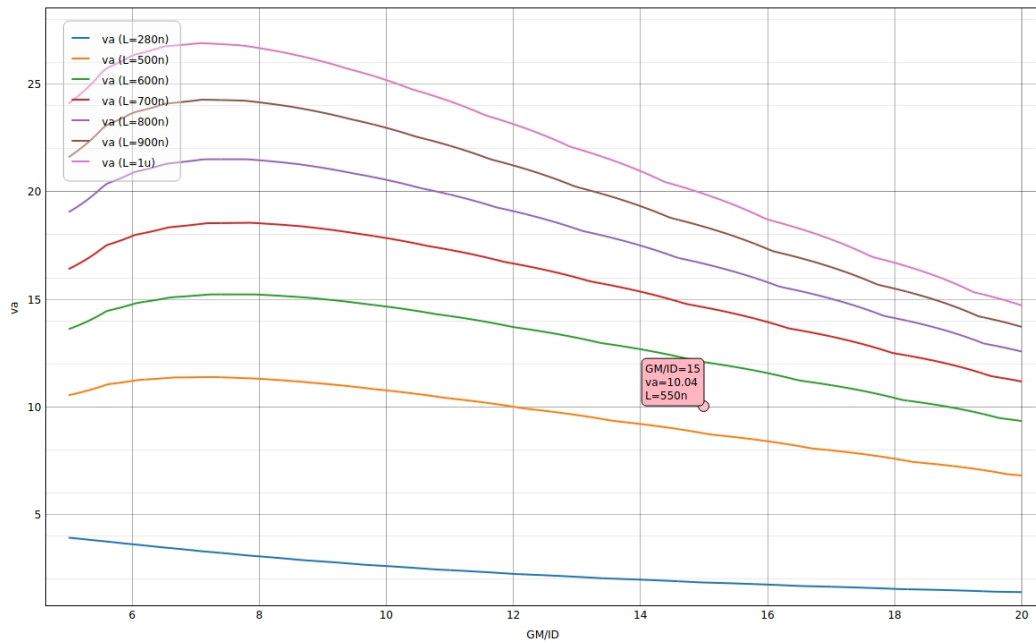
II. First-Stage Current Mirror Load (M6 / M1)

Step 9) Compute the required VA (assuming $gm/ID = 15$ as an initial guess):

$$(g_m/I_D)_{assumed} = 15$$

$$V_A = I_D / g_{ds} = 6.25 \mu A / 0.623 \mu S \approx 10 V$$

From the charts at $V_A \approx 10 V$: $L = 550 \text{ nm}$

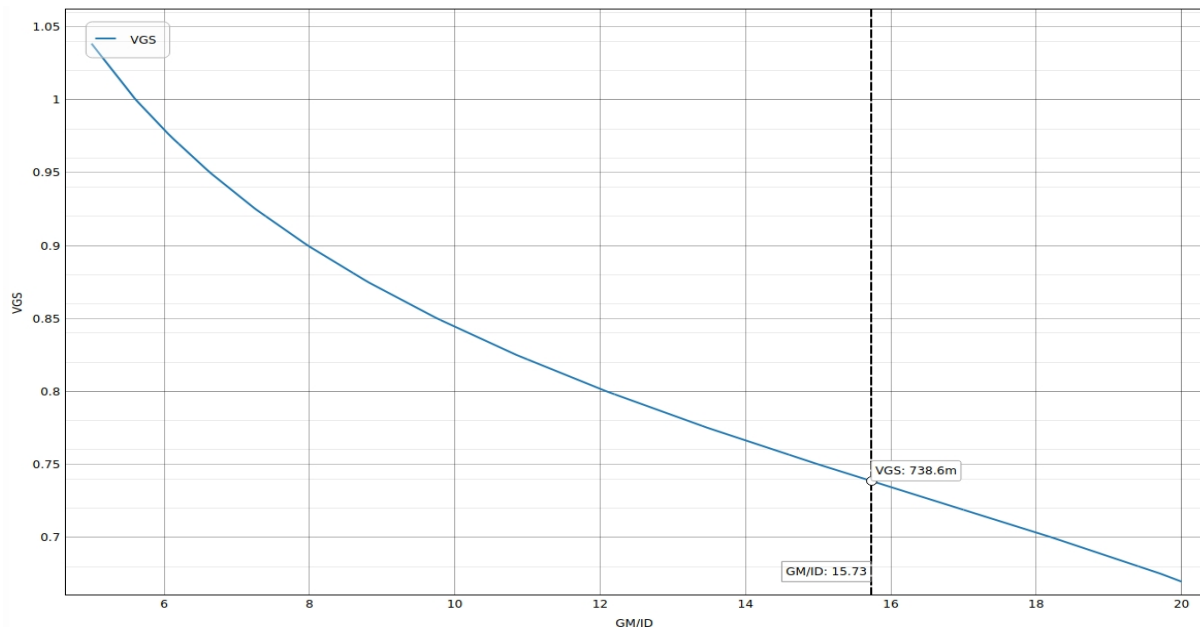


Step 15) Match VGS with the second stage (to prevent systematic offset):

note that this step will get after next steps

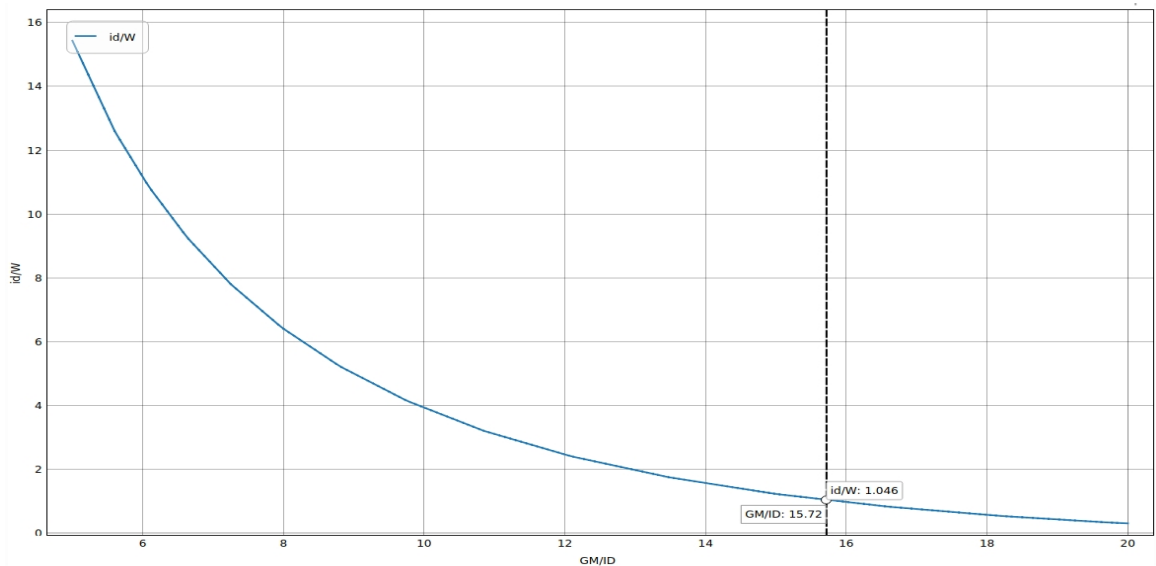
$$VGS(\text{mirror load}) = VGS(M2, \text{second stage})$$

From the VGS chart at $L = 550 \text{ nm}$, the value that satisfies the match:



$$(g_m/I_D)_{load,final} = 15.73$$

Compute W:



$$W_{6,1} = I_D / (I_D/W)_{\text{chart}} = 6.25 \mu A / 1.045 \approx 5.98 \mu m$$

III. Second-Stage Input Transistor (M2)

Step 10) Compute gm_2 from the Phase Margin specification (assuming $\omega_{p2} = 4\omega_u$):

$$\omega_{p2} = g_{m2} / (2\pi C_L) = 4\omega_u = 20 \text{ MHz}$$

$$g_{m2} \approx 630 \mu S$$

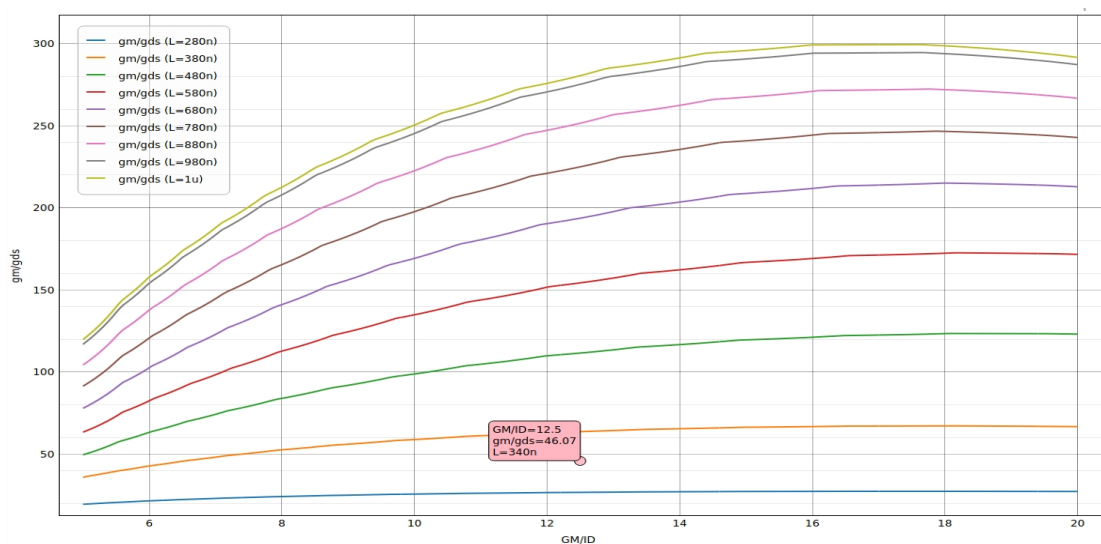
$$(g_m/I_D)_2 = 630 / 47.5 = 13.3$$

Step 14) Compute g_{ds} and L for the second-stage transistor:

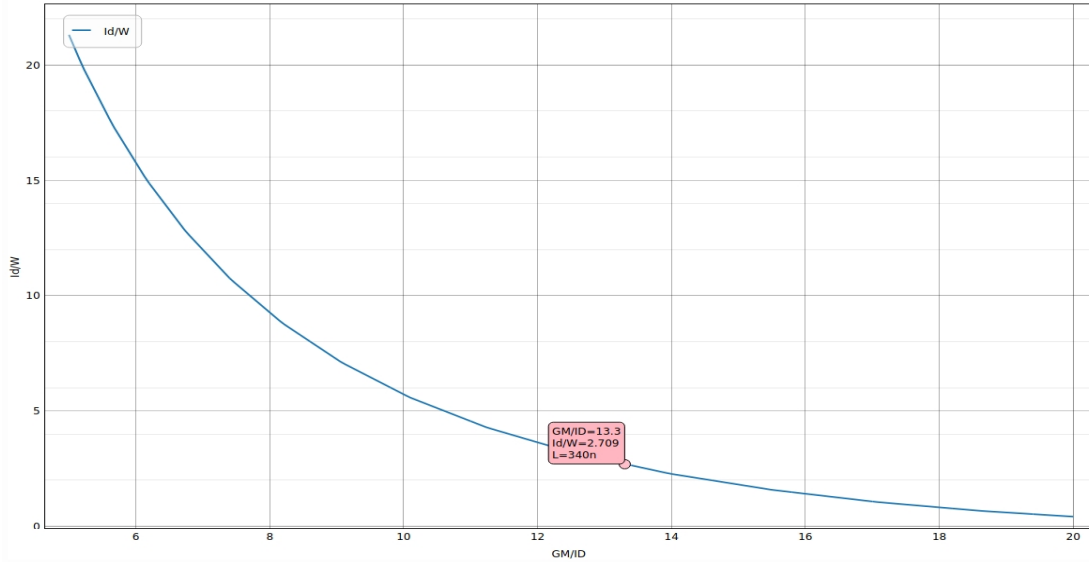
$$A_{v2} = g_{m2} / (g_{ds,in2} + g_{ds,load})$$

$$g_{ds,in2} = g_{m2} / A_{v2} - g_{ds,load} \approx 13.95 \mu S$$

$$(g_m/g_{ds})_{\text{target}} = 630 / 13.95 \approx 45.2$$



From the charts: $L = 340 \text{ nm}$



Compute W and V_{GS} :

$$W_2 = 47.5 \mu A / 2.7 \approx 17.59 \mu m$$

$$V_{GS,2} = 738.9 \text{ mV}$$

This value V_{GS} Step 15 above to match V_{GS} with the mirror load.

IV. Tail Current Source (M4) and Second-Stage Load (M3)

Step 11) Compute v_{dsat} and gm/ID (from the Swing-high and CMIR-high specifications):

Swing-high:

$$V_{out,max} = V_{DD} - V_{ov,load2} = 1.8 - 0.2 = 1.6 \text{ V} \Rightarrow V_{ov,max} \leq 0.2 \text{ V}$$

$$v_{dsat,tail} < V_{DD} - V_{GS}(M8) - V_{CM,high} \Rightarrow v_{dsat,tail,max} \approx 0.118 \text{ V}$$

Taking the smaller value:

$$V^* = v_{dsat} = 0.118 \text{ V}$$

$$(g_m/I_D)_{tail,load2} = 2 / 0.118 \approx 17$$

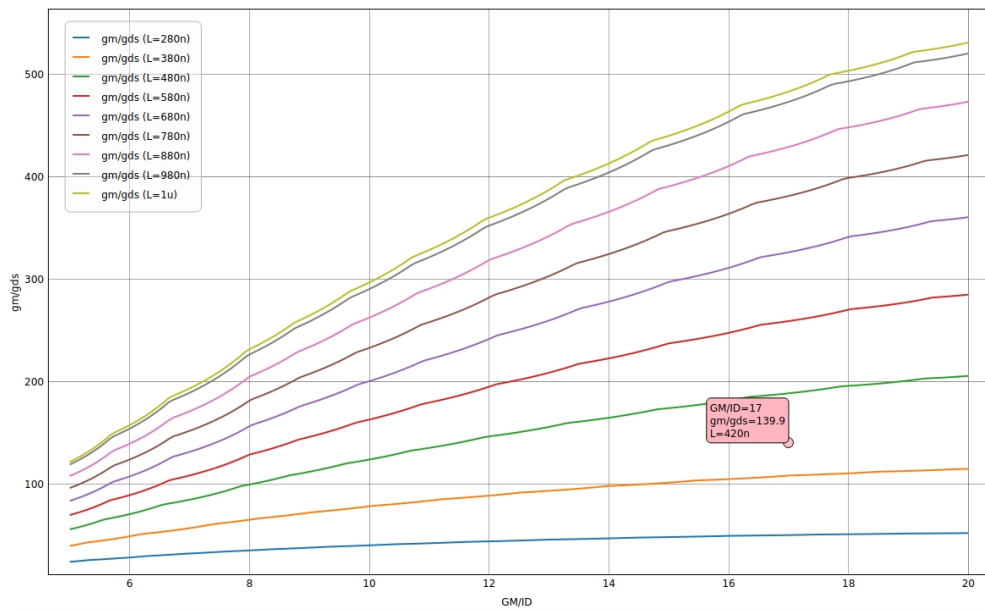
Step 12) Compute $g_{ds,tail}$ from the CMRR specification:

$$A_{vcm} = -g_{ds,tail} / (2 \cdot g_{m,load(mirror)})$$

$$CMRR = A_{v1} - A_{vcm} \Rightarrow A_{vcm} = 36 - 74 = -38 \text{ dB}$$

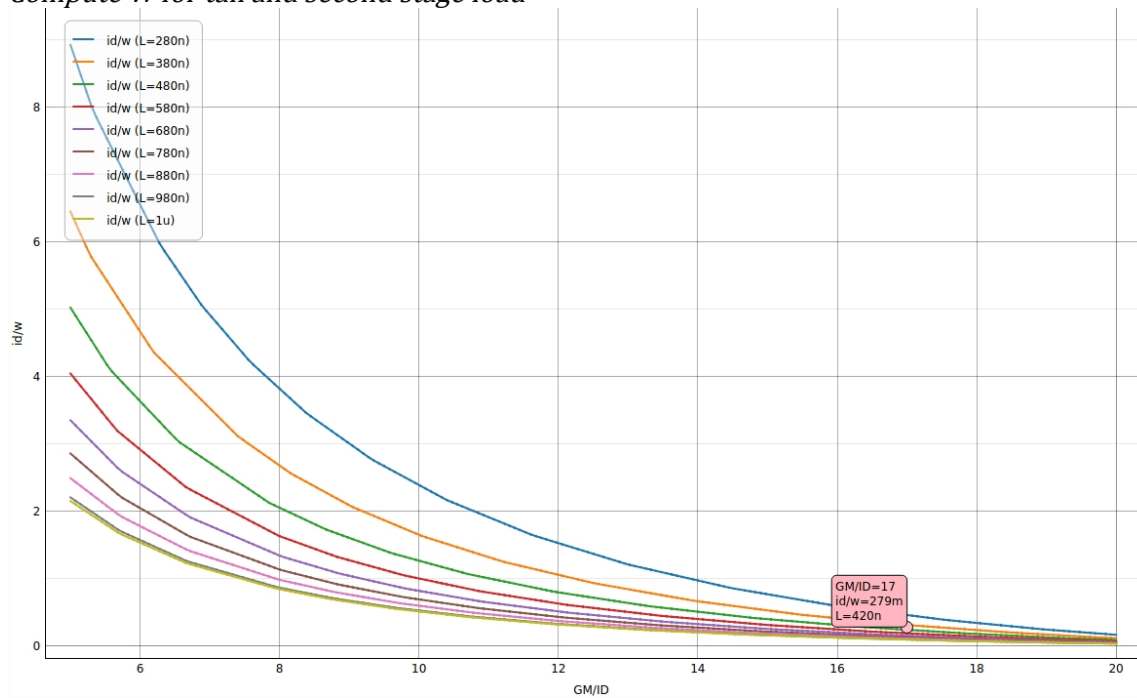
$$g_{ds,tail} \leq 1.5725 \mu S$$

$$g_m/g_{ds} = 17 (12.5) / 1.57u = 135.3$$



From chart $L = 420 \text{ nm}$

Compute W for tail and second stage load



$Id/w = 279 \text{ m}$

$W_{tail} = id / (id/w) = 48 \text{ u}$

$W_{Load_Second_Stage} = Id / (id/w) = 170 \text{ u}$

After tuning the nmos load transistor to keep its gate voltage equal to VGS of second stage and keep Vout equal to vdd/2

L= 425 nm and W = 4.2u

Also Cc = 2.2 pF (to verify settling time and GBW)

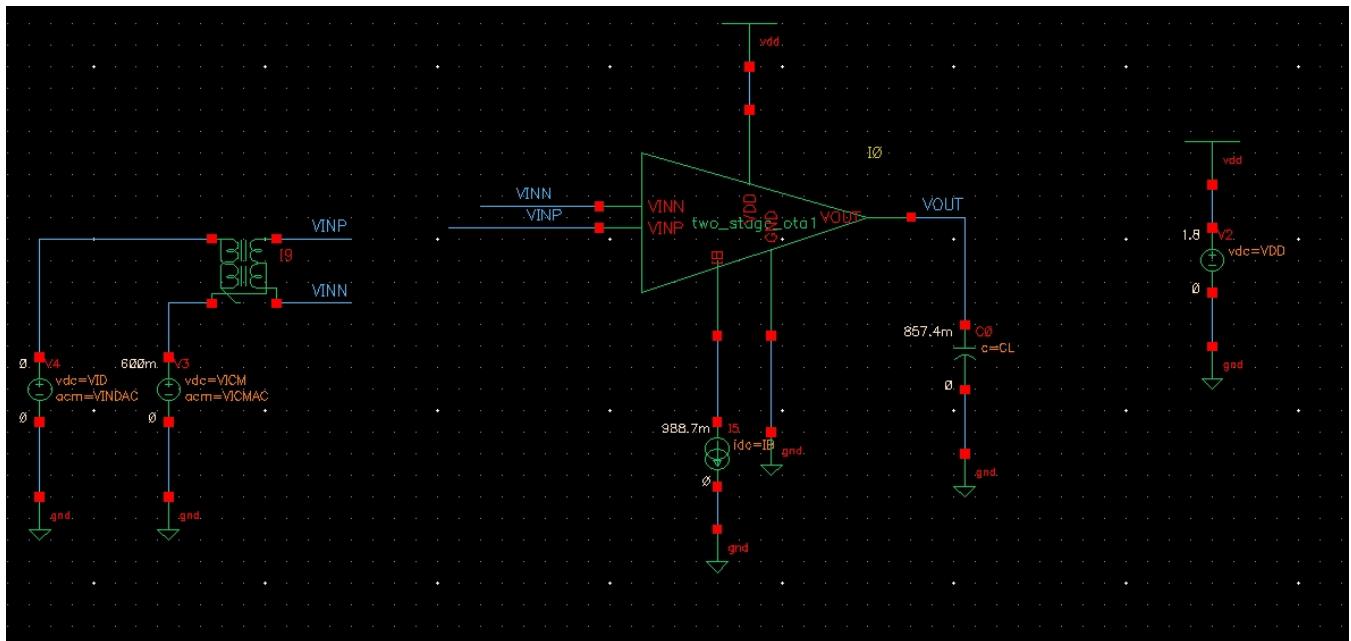
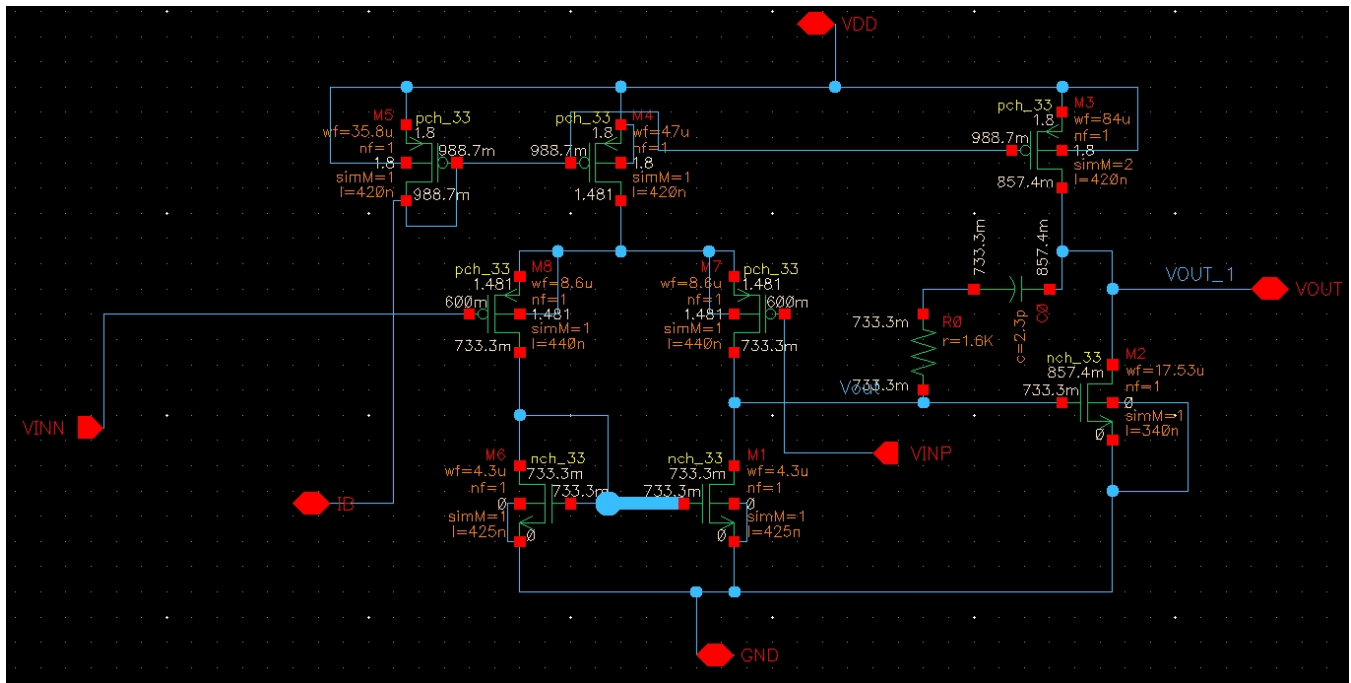
$R_z = 1 / g_{m2} = 1.6 \text{ k}\Omega$

V. Sizing Summary Table

Transistor	Role	ID (μA)	W (μm)	L (nm)	gm (μA/V)	gm/ID (V ⁻¹)	vdsat (V)	Vov (V)	V* (V)
M7, M8	1st-stage input pair	6.25	8.5u	440n	78.5	12.6	134.3m	100m	158.4m
M6, M1	1st-stage mirror load	6.25	4.3u	420n	93.19u	15.04	105.2m	35.4	133m
M4	Tail current source	12.5	48u	420n	212.4u	17	89.53m	26.92m	117.7
M2	2nd-stage input	47.5	17.53u	340n	628.3	13.2	120m	62.56m	153.1m
M3	2nd-stage current-source load	47.5	170u	420n	792.9u	17	91.92	31.32m	119.8m

Part 3 — Open-Loop OTA Simulation

3.1 Schematic, bias circuit, and DC operating point



Transistor	Role	ID (μA)	gm (μS)	gds (S)	gm/ID (V^{-1})	vdsat (mV)	Vgs (mV)	V* (V)
M7, M8	1st-stage input pair (PMOS)	6.17	80.09	525.8 n	12.6	130	880.5	—
M1, M6	1st-stage mirror load	6.17	94.22	925 n	—	104	733.3	—
M4	Tail current source	12.35	211.4	2 u	—	90	811.3	—
M2	2nd-stage input (common source)	47.53	631u	11 u	13.2	117.9	733.3	—
M3	2nd-stage current-source load	47.53	808u	4.6 u	—	90	811.2	—
M5	10 μA reference $\rightarrow$ IB1, I2	10	170.3	1.029 u	—	90	811.3	—

All transistor operate in saturation

Is the current (and gm) in the input pair exactly equal?

With VIDAC = 0 (zero differential input) and a perfectly symmetric layout, ID(M1) and ID(M2) are ideally equal by circuit symmetry. In practice.

What is the DC voltage at the output of the first stage? Why? (733 m)

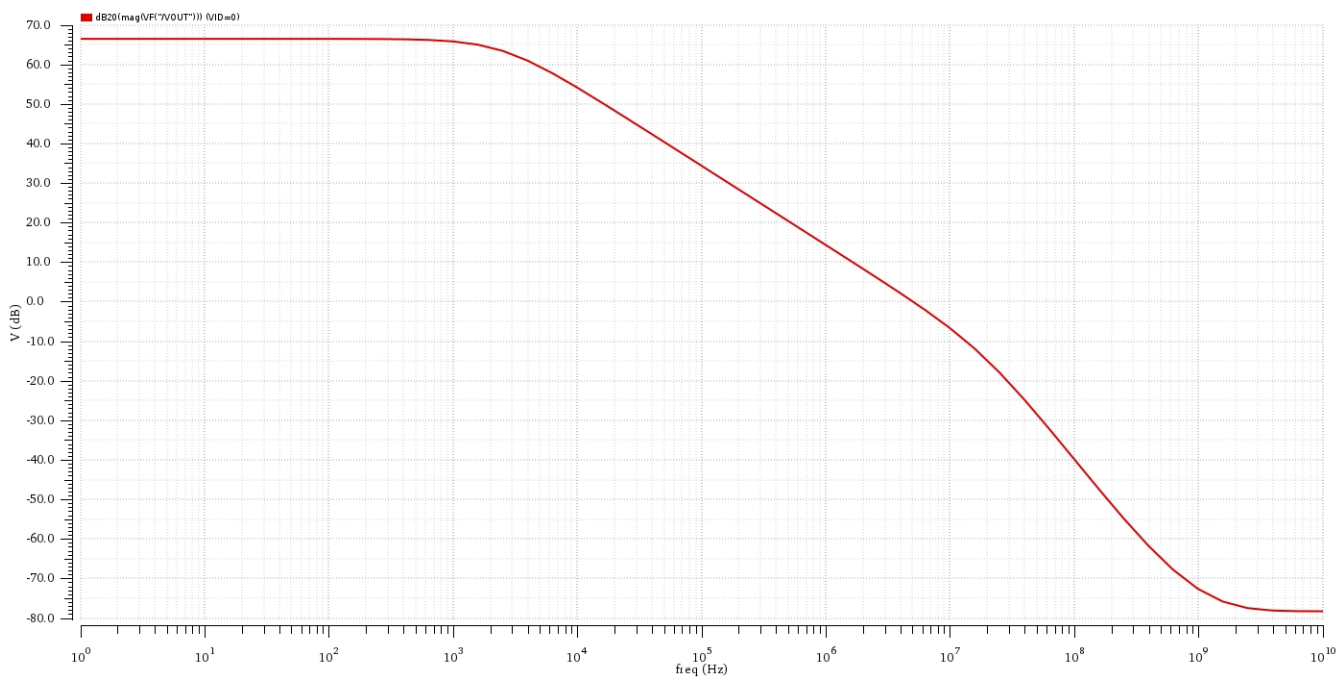
It is equal to 733 m and it equal to vgs of second stage and mirror load of first stage
we have done a sweep on W of mirror load to cancel systematic offset

What is the DC voltage at the output of the second stage? Why?(857)

It is equal to 857m and it is near to VDD/2
because at zero offset the output will be at when two current of mosfets balanced, and we sizing them to work at vdd/2 to maximize the output swing.

3.2 Differential small-signal AC

diff-gain-vs-frequency



my_labs:two_stage_tb:1	Ao	2.129k			
my_labs:two_stage_tb:1	Ao_dB	66.56			
my_labs:two_stage_tb:1	BW	2.49k			
my_labs:two_stage_tb:1	fu	5.391M			
my_labs:two_stage_tb:1	(Ao * BW)	5.3M			

Hand calc

$A_o = g_{m7}(r_{o7} || r_{o1}) * g_{m2}(r_{o2} || r_{o6}) = 2228 \text{ (66.96 dB)}$

$BW = 1/2\pi C_c(AV_2)R_{out1} = 2.589 \text{ kHz}$

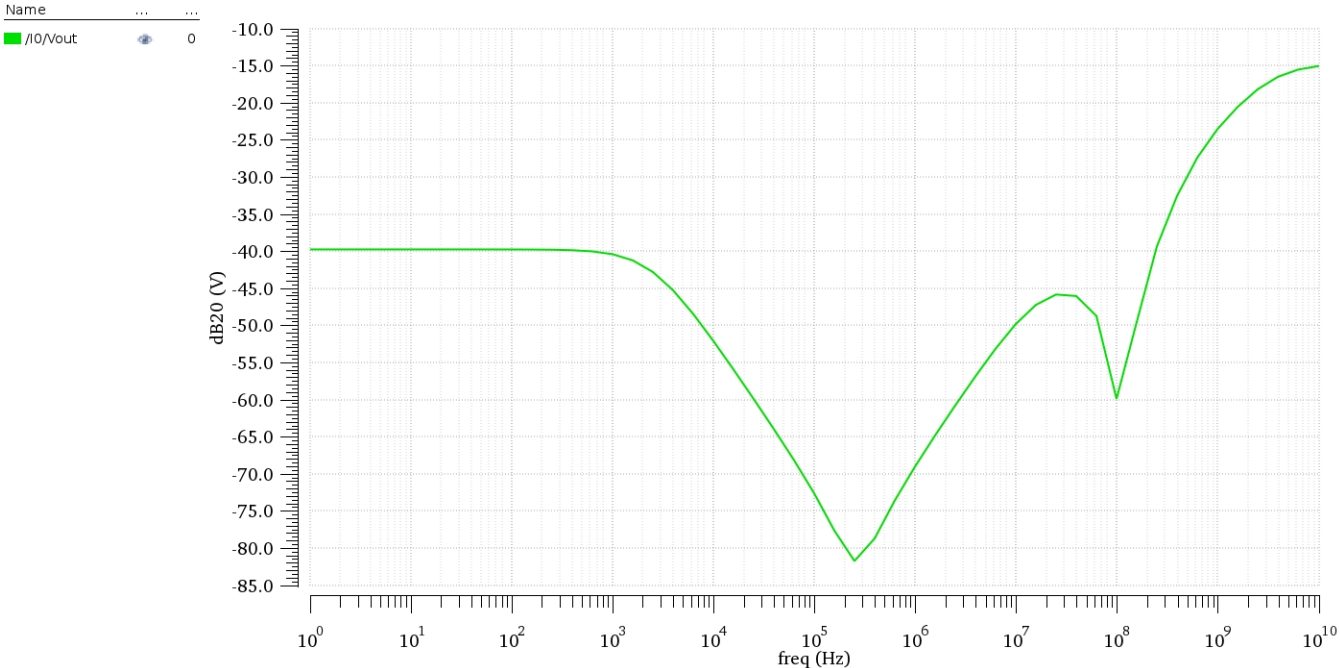
$GBW = A_o * BW = 5.7 \text{ MHz}$

Parameter	Hand calculation	Simulation
Ao (dB)	2.228 k (66.96 dB)	2.129 (66.56 dB)
BW (Hz)	2.589 k	2.49k
GBW	5.7 M	5.3 MHz

$\text{Error} = 1/A_{oI} * 100 = .044\% < 0.05\%$

3.3 Common-mode small-signal AC

CM-gain-vs-frequency

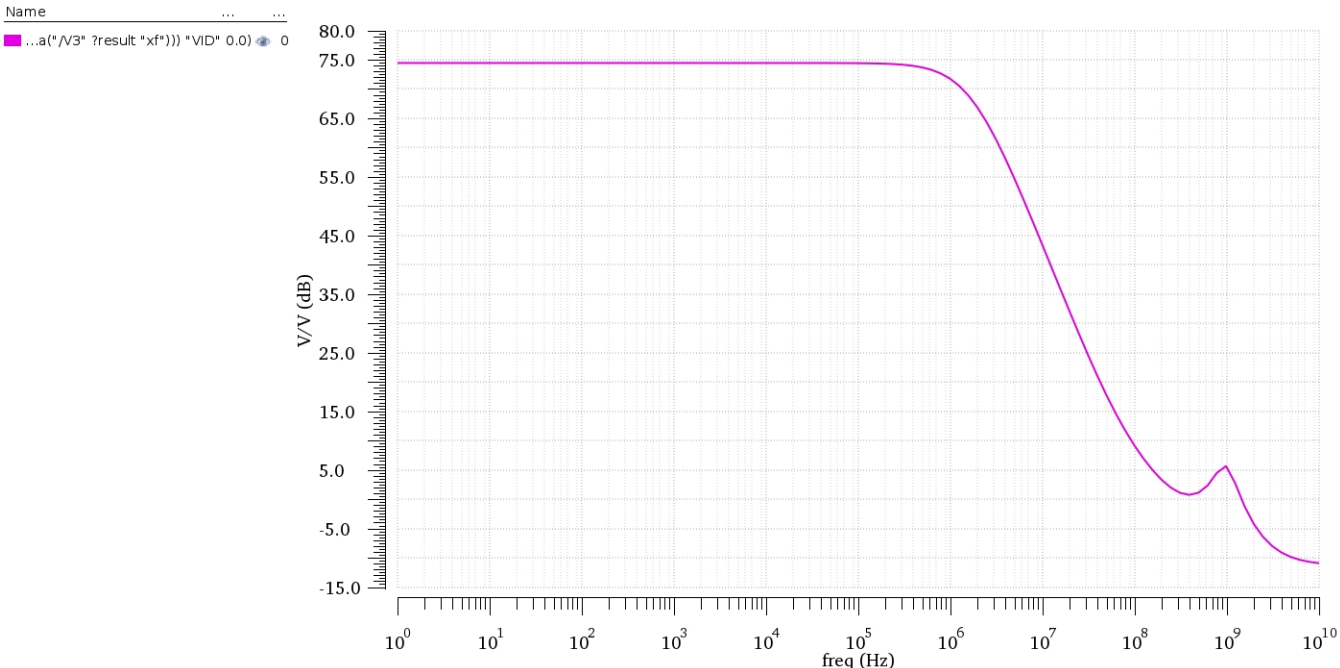


AVCM = gds_tail / 2gmlload = -39.48 dB
AVCM is only for first stage .

Parameter	Hand calculation	Simulation
Acm(dc) (dB)	-39.5 dB	-39.7 dB

3.4 CMRR (optional)

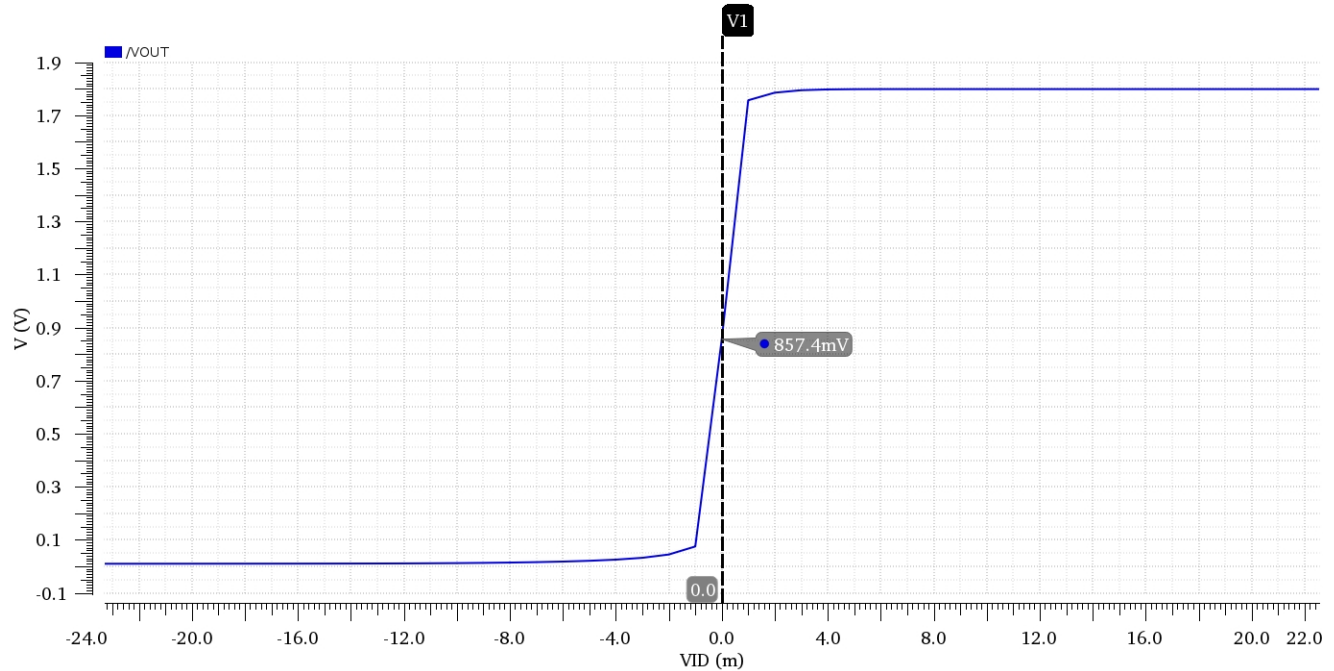
CMRR-vs-frequency plot



Parameter	Hand calculation	Simulation
CMRR @ DC (dB)	74 dB (target)	74.5 dB

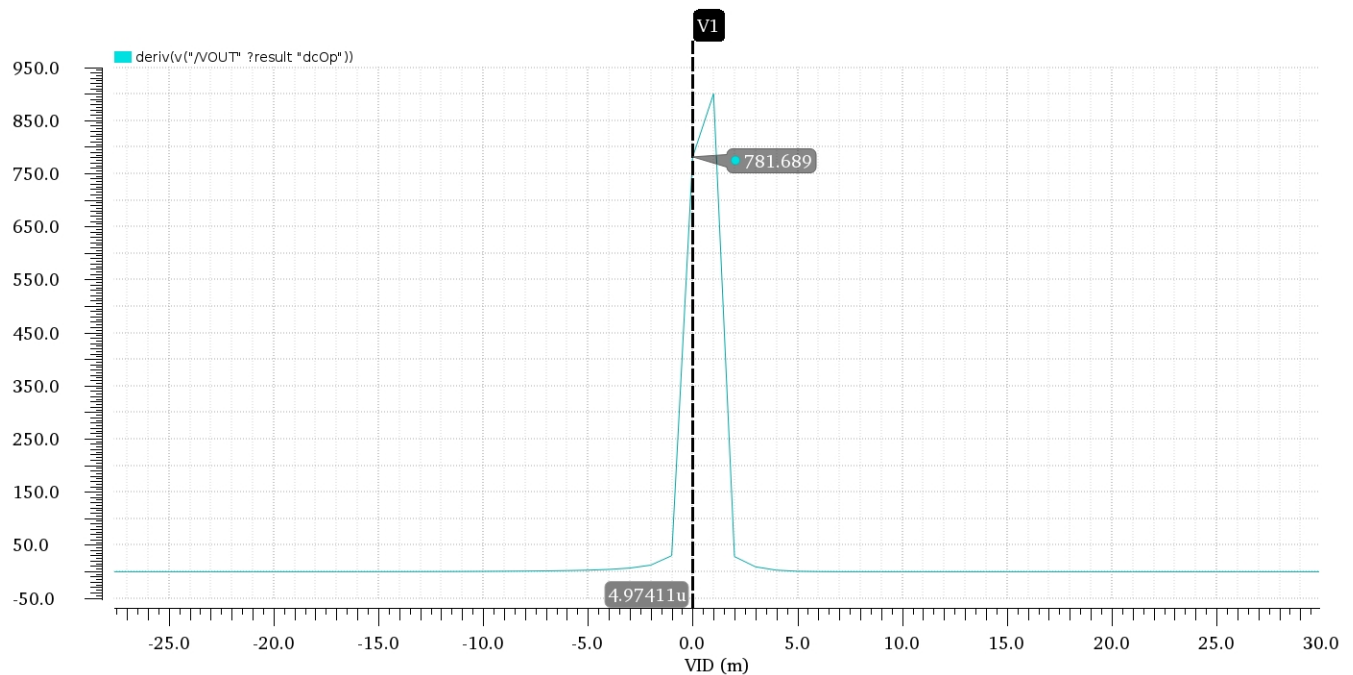
3.5 Diff large signal ccs (optional)

VOUT-vs-VID



VOUT at 0 is 857 is equal to vout at dc op simulation ,

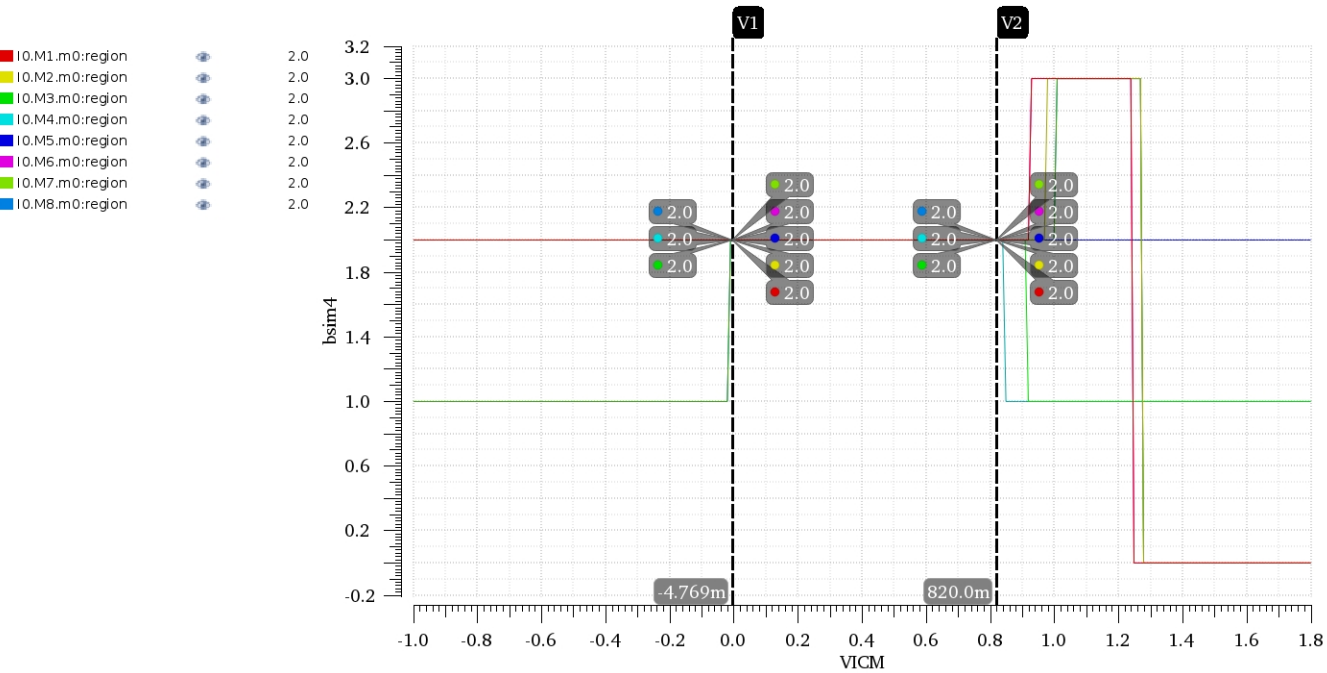
Derivative VOUT-vs-VID



The peak value of dervative at 0 V is much less than dc gain (2000) ,since it use finnite voltage step but to get the right value we need infinity number of simulation points

3.6 Common-mode large signal

Region -vs Vincm



hand analysis:

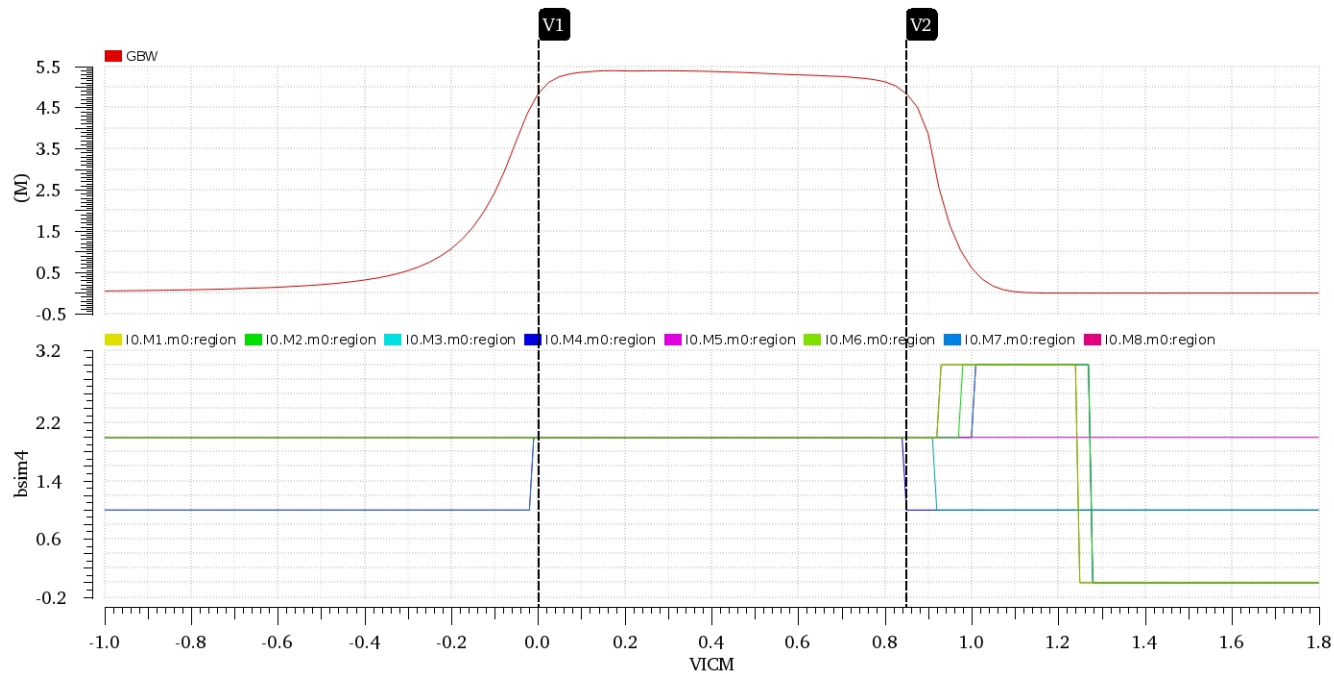
$VCMR_{low} = VDD - VGS7 - VDSsat4$

$VCMR_{high} = VGS7 - vdsat7 - VGS1$

Parameter	Hand calculation	Simulation
VCMR_low	-4 m	0
VCMR_High	800m	820m

3.7 Common-mode large signal (optional)

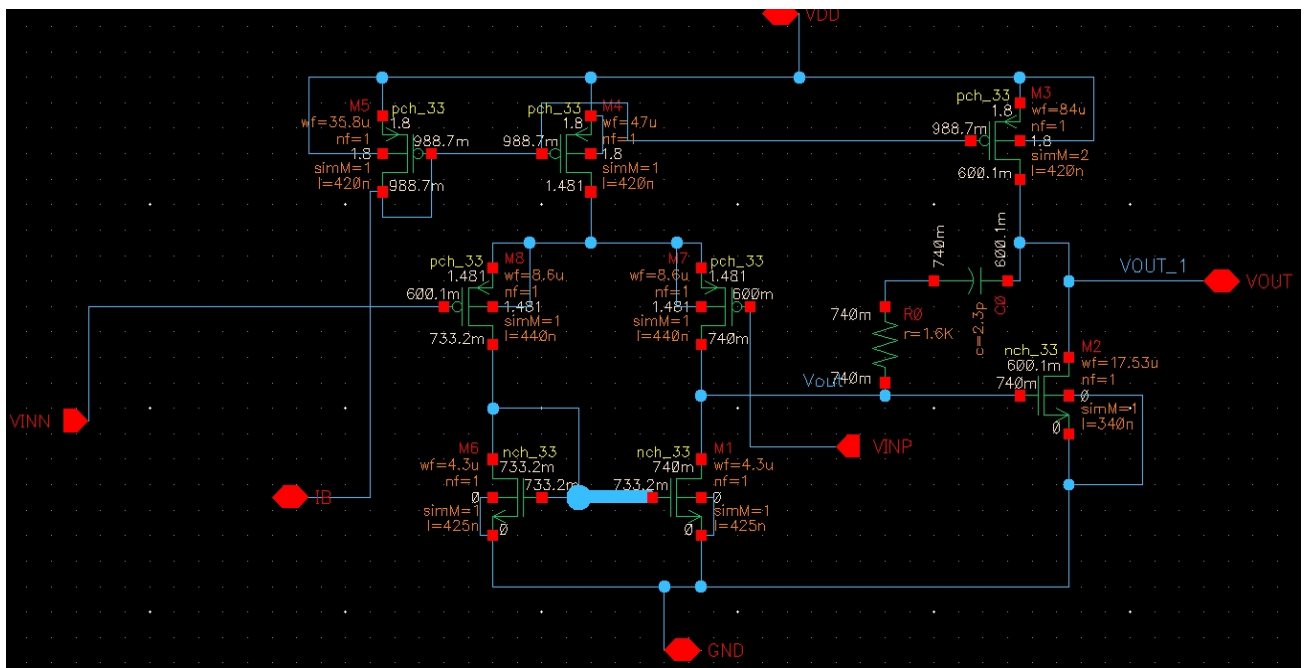
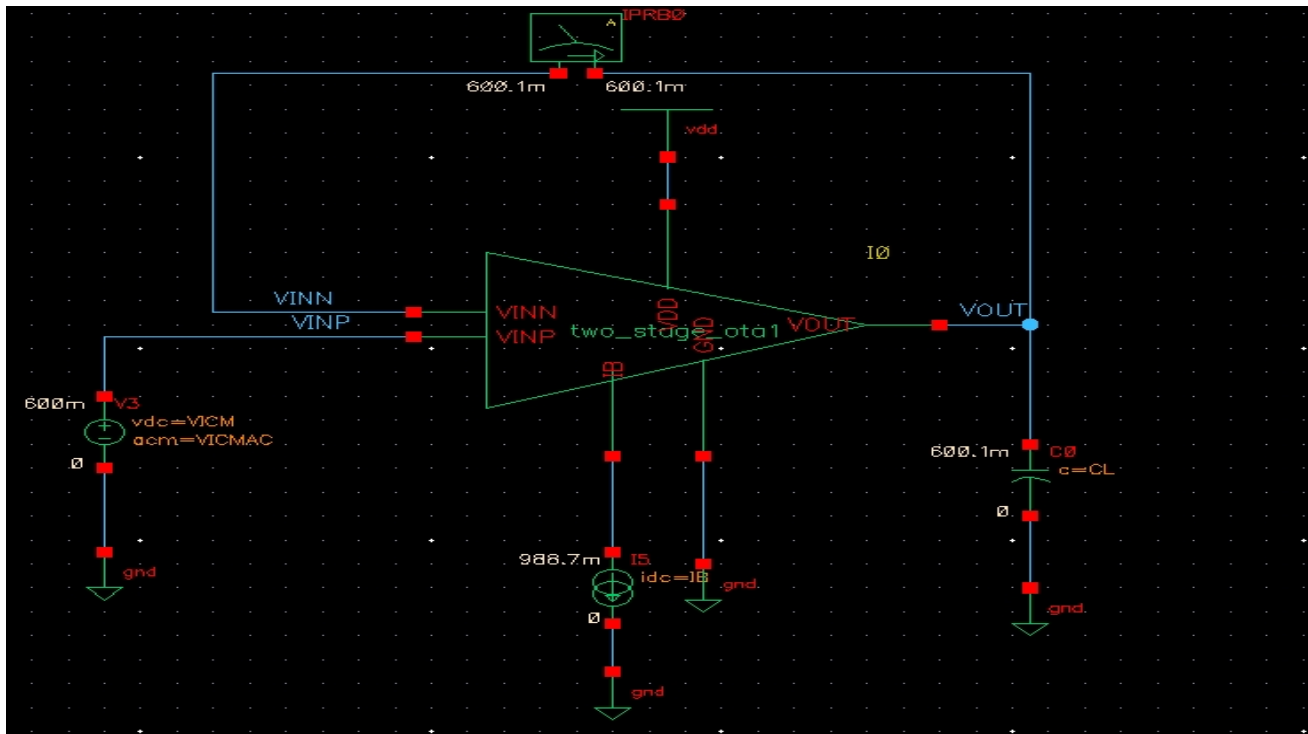
GBW -vs- Vincm



Parameter	Simulation
VCMR_low (90% 5.4MHz)	0
VCMR_High (90% 5.4MHz)	830m

Part 4 — Closed-Loop OTA Simulation (Unity-Gain Buffer)

4.1 Schematic and DC operating point



my_labs:two_stage_tb:1	VINN	600.1m			
my_labs:two_stage_tb:1	VINP	600m			
my_labs:two_stage_tb:1	VOUT	600.1m			
my_labs:two_stage_tb:1	Vout-1 stage	740m			

	Role	ID (μA)	gm (μS)
M7	1st-stage input pair	6.179	80.11
M8	1st-stage input pair	6.173	80.09

Are the DC voltages at the input terminals of the op-amp exactly equal? Why?

Not exactly in an op-amp with negative feedback the two input terminals are forced very close together but not equal, the difference between them is the static error and that because op amp have finite gain.

Is the DC voltage at the output of the first stage exactly equal to the open-loop value? Why?

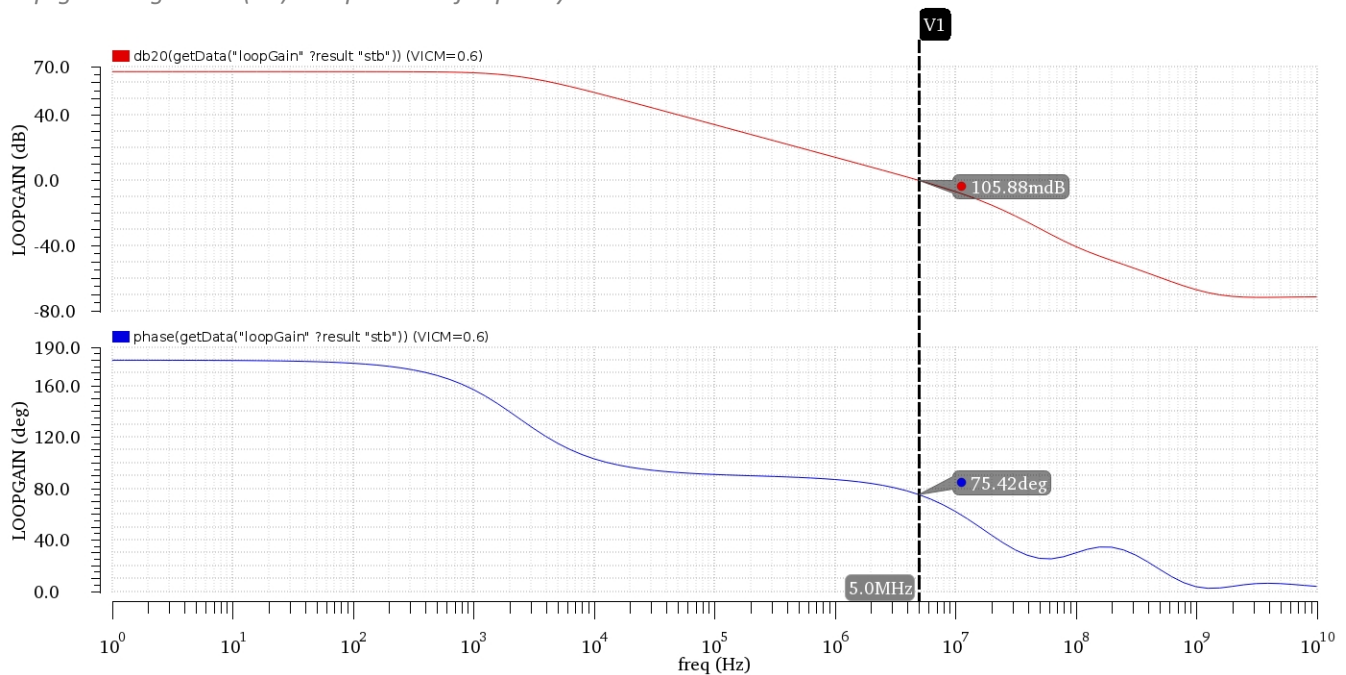
No they're close 740 and 733m but not exactly equal, because in closed loop the feedback forces $V_{OUT} = V_{IN}$ exactly, which shifts the output first stage to be compared with open loop value.

Is the current (and gm) in the input pair exactly equal? Why?

No — there are very small current/gm imbalance (μA -level) exists between M1 and M2, because the loop needs to force $V_{OUT} = V_{IN}$, so there must be an offset voltage between input pair.

4.2 Loop gain (STB analysis)

loop-gain magnitude (dB) and phase vs. frequency



LG	66.89			
UGF	5.067M			
GBW	5.246M			
PM	75.2			

Hand analysis

$$DC\ LG = B \cdot A_{OL} = A_{OL} = 66.96\ dB$$

$$W_u = GBW = g_m / 2 \cdot \pi \cdot C_c = 5.305\ MHz$$

$$W_{p2} = 4 \cdot w_u = 31.22\ MHz$$

$$PM = 90 - \tan^{-1}(w_u / W_{p2}) = 76$$

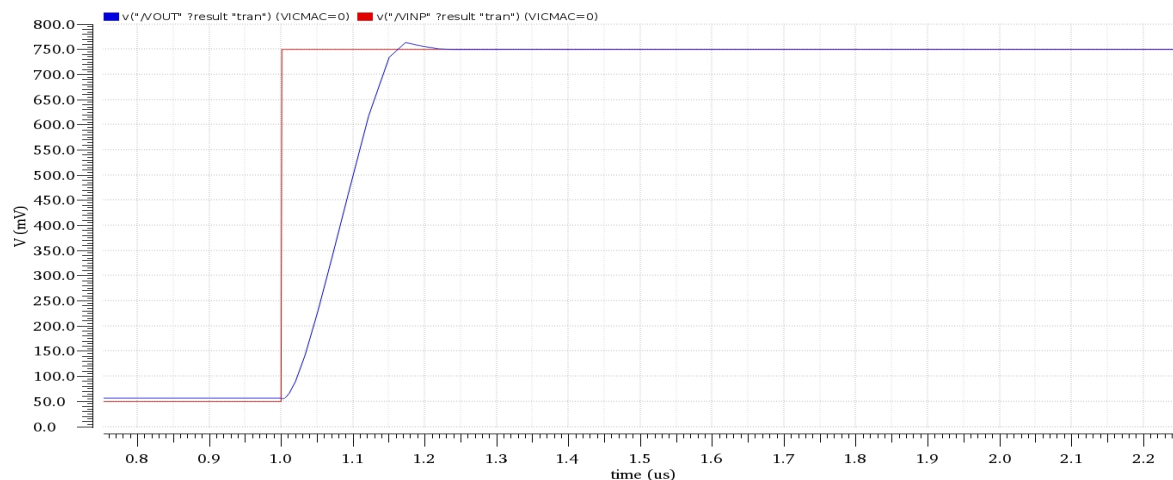
Parameter	STB sim (closed loop)	Hand calculation	Open-loop sim
DC gain (dB)	66.89	66.96	66.56
fu (Hz)	5.067 M	5.305 M	5.3 M
GBW (Hz)	5.246 M	5.305 M	5.39 M
Phase margin (°)	75.2	76	-----

The STB closed-loop results showt agreement with the hand calculations and open-loop simulation with only error less than 5%

a small differene in phase margin that because we have decrease compensation cap.

4.3 Slew rate

Vin/Vout overlay plot



Hand Analysis :

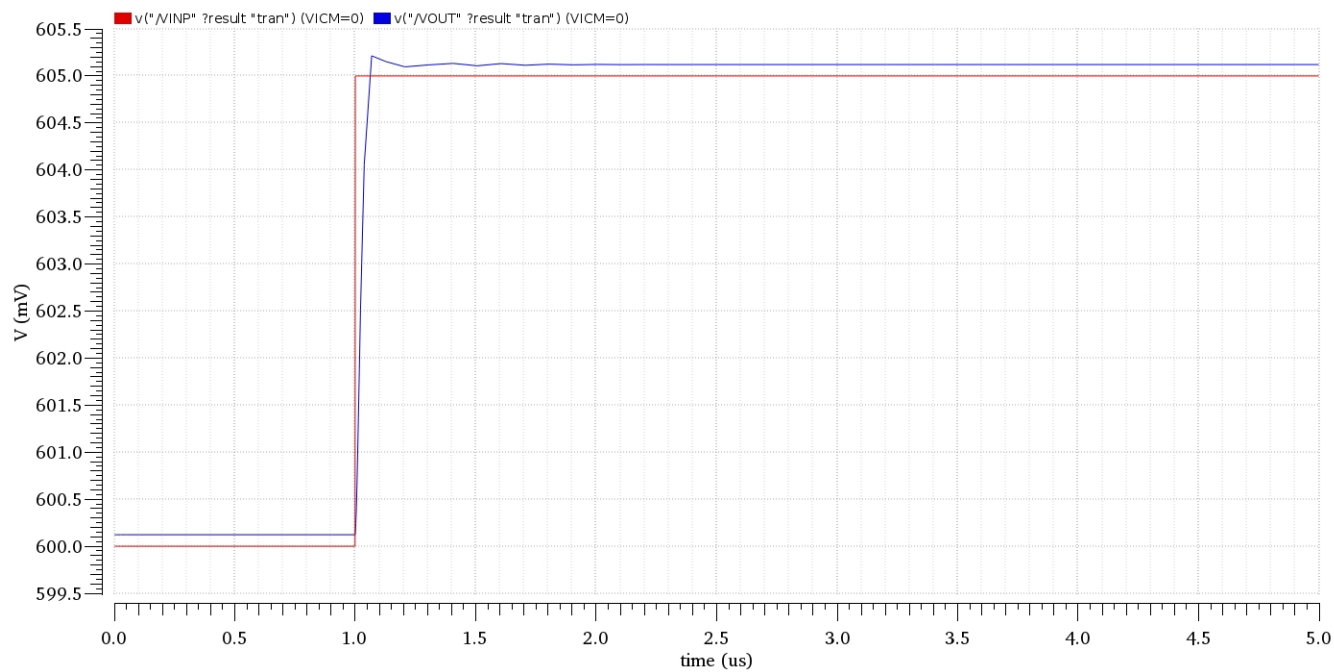
$$Slew\ rate = IB1 / CC = 12.5u / 2.2p = 5.68\ (V/uS)$$

slew_rate	5.132M
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Parameter	Hand calculation	Simulation
Slew rate (V/μs)	5.68	5.132

4.4 Settling time

Transient plot



my_labs:two_stage_tb:1	rise	43.46n
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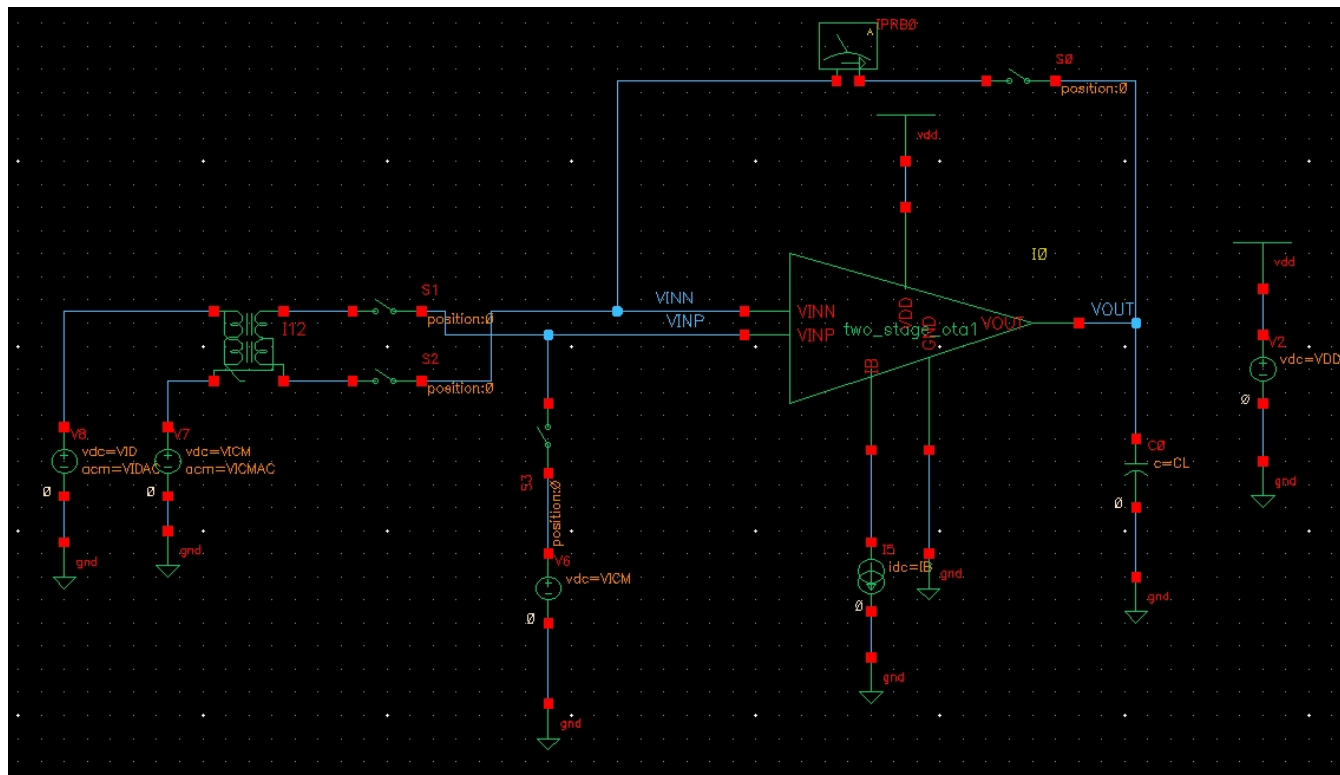
Hand analysis

rise time = $2.2\tau = 2.2 * 1/W_u = 69.1 \text{ n}$

Parameter	Hand calculation (2.2τ)	Simulation	% error
Rise time (ns)	69.1n	43.46 n	

The simulated rise time is typically faster than the 2.2τ hand estimate. This is because 2.2τ is a first order , but the real closed-loop OTA is a second order system and 75.2 phase margin the closed-loop step response is cricital damped and rise time is faster than expected, and of there is some ringing because phase margin is less that 76 .

Part 5 (Optional) — DC-Closed-Loop / AC-Open-Loop Simulation



5.1 Dc operation Point

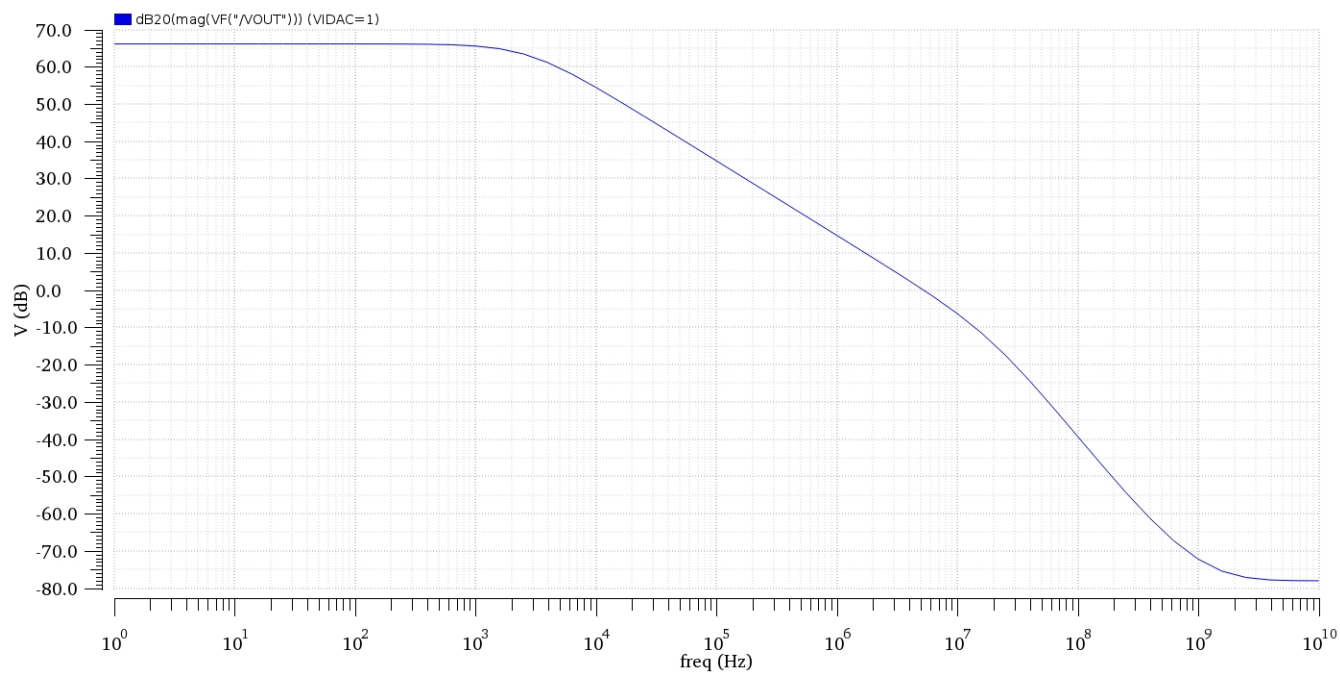
my_labs:two_stage_tb:1	VinN	600.1m
my_labs:two_stage_tb:1	VOUT	600.1m
my_labs:two_stage_tb:1	Vout1 stage	740m
my_labs:two_stage_tb:1	pv("I0.M1.m0" "id" ?result "dc...	6.179u
my_labs:two_stage_tb:1	pv("I0.M2.m0" "id" ?result "dc...	48.68u
my_labs:two_stage_tb:1	pv("I0.M3.m0" "id" ?result "dc...	-48.68u
my_labs:two_stage_tb:1	pv("I0.M4.m0" "id" ?result "dc...	-12.35u
my_labs:two_stage_tb:1	pv("I0.M5.m0" "id" ?result "dc...	-10u
my_labs:two_stage_tb:1	pv("I0.M6.m0" "id" ?result "dc...	6.173u
my_labs:two_stage_tb:1	pv("I0.M7.m0" "id" ?result "dc...	-6.179u
my_labs:two_stage_tb:1	pv("I0.M8.m0" "id" ?result "dc...	-6.173u
my_labs:two_stage_tb:1	pv("I0.M1.m0" "gm" ?result "...	94.26u
my_labs:two_stage_tb:1	pv("I0.M2.m0" "gm" ?result "...	640.5u
my_labs:two_stage_tb:1	pv("I0.M3.m0" "gm" ?result "...	824.8u
my_labs:two_stage_tb:1	pv("I0.M4.m0" "gm" ?result "...	211.4u
my_labs:two_stage_tb:1	pv("I0.M5.m0" "gm" ?result "...	170.3u
my_labs:two_stage_tb:1	pv("I0.M6.m0" "gm" ?result "...	94.18u
my_labs:two_stage_tb:1	pv("I0.M7.m0" "gm" ?result "...	80.11u
my_labs:two_stage_tb:1	pv("I0.M8.m0" "gm" ?result "...	80.06u
my_labs:two_stage_tb:1	(pv("I0.M1.m0" "gm" ?result "...	15.25
my_labs:two_stage_tb:1	(pv("I0.M7.m0" "gm" ?result "...	-12.96
my_labs:two_stage_tb:1	(pv("I0.M2.m0" "gm" ?result "...	13.16
my_labs:two_stage_tb:1	(pv("I0.M4.m0" "gm" ?result "...	-17.11
my_labs:two_stage_tb:1	pv("I0.M1.m0" "region" ?resu...	2
my_labs:two_stage_tb:1	pv("I0.M2.m0" "region" ?resu...	2
my_labs:two_stage_tb:1	pv("I0.M3.m0" "region" ?resu...	2
my_labs:two_stage_tb:1	pv("I0.M4.m0" "region" ?resu...	2
my_labs:two_stage_tb:1	pv("I0.M5.m0" "region" ?resu...	2
my_labs:two_stage_tb:1	pv("I0.M6.m0" "region" ?resu...	2
my_labs:two_stage_tb:1	pv("I0.M7.m0" "region" ?resu...	2
my_labs:two_stage_tb:1	VinP	600m

All transistor in saturation region

Output voltage of first stage 740m

the output voltage of Amp is set by negative feedback network

5.1 Diff Ac small signal



my_labs:two_stage_tb:1	Ao	2.051k			
my_labs:two_stage_tb:1	Ao_dB	66.24			
my_labs:two_stage_tb:1	BW	2.724k			
my_labs:two_stage_tb:1	fu	5.591M			
my_labs:two_stage_tb:1	(Ao * BW)	5.587M			
my_labs:two_stage_tb:1	PM	75.4			

Parameter	Open-loop sim	Closed loop/ open Loop
DC gain (dB)	66.56	66.24
fu (Hz)	5.3 M	5.591 M
GBW (Hz)	5.39 M	5.587 M